Prefix vs. Postfix

- `++` and `--` operators can be used in complex statements and expressions
- In prefix mode (`++val, --val`) the operator increments or decrements, *then* returns the value of the variable
- In postfix mode (`val++, val--`) the operator returns the value of the variable, *then* increments or decrements

Prefix vs. Postfix - Examples

```
int num, val = 12;
cout << val++; // displays 12,
               // val is now 13;
cout << ++val; // sets val to 14,
                // then displays it
num = --val;   // sets val to 13,
               // stores 13 in num
num = val--;   // stores 13 in num,
               // sets val to 12
```

Notes on Increment and Decrement

- Can be used in expressions:

```
result = num1++ + --num2;
```

- Must be applied to something that has a location in memory. Cannot have:

```
result = (num1 + num2)++;
```

- Can be used in relational expressions:

```
if (++num > limit)
```

pre- and post-operations will cause different comparisons

Watch Out for Infinite Loops

- The loop must contain code to make *`expression`* become `false`
- Otherwise, the loop will have no way of stopping
- Such a loop is called an *infinite loop*, because it will repeat an infinite number of times

Using the `while` Loop for Input Validation

- Input validation is the process of inspecting data that is given to the program as input and determining whether it is valid.
- The while loop can be used to create input routines that reject invalid data, and repeat until valid data is entered.

Using the `while` Loop for Input Validation

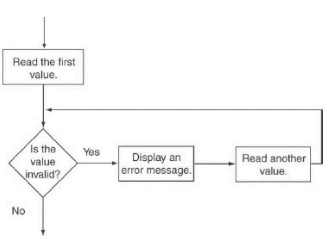
- Here's the general approach, in pseudocode:

Read an item of input.
While the input is invalid
Display an error message.
Read the input again.
End While

Input Validation Example

```
cout << "Enter a number less than 10: ";
cin >> number;
while (number >= 10)
{
   cout << "Invalid Entry!"
        << "Enter a number less than 10: ";
   cin >> number;
}
```

Flowchart for Input Validation



Counters

- Counter: a variable that is incremented or decremented each time a loop repeats
- Can be used to control execution of the loop (also known as the *loop control variable*)
- Must be initialized before entering loop

The `do-while` Loop

- `do-while`: a posttest loop – execute the loop, then test the `expression`
- General Format:

```
do
   statement;  // or block in { }
while (expression);
```

- Note that a semicolon is required after (*expression*)

`do-while` Loop Notes

- Loop always executes at least once
- Execution continues as long as *`expression`* is `true`, stops repetition when *`expression`* becomes `false`
- Useful in menu-driven programs to bring user back to menu to make another choice (*see* Program 5-8 *on pages* 245-246)

The `for` Loop

- Useful for counter-controlled loop
- General Format:

```
for(initialization; test; update)
    statement; // or block in { }
```

- No semicolon after the `update` expression or after the)

`for` Loop - Mechanics

```
for(initialization; test; update)
        statement; // or block in { }
```

1) Perform *`initialization`*
2) Evaluate *`test`* expression
   - If `true`, execute *`statement`*
   - If `false`, terminate loop execution
3) Execute *`update`*, then re-evaluate *`test`* expression

When to Use the `for` Loop

- In any situation that clearly requires
  - an initialization
  - a false condition to stop the loop
  - an update to occur at the end of each iteration

The `for` Loop is a Pretest Loop

- The for loop tests its test expression before each iteration, so it is a pretest loop.
- The following loop will never iterate:

```
for (count = 11; count <= 10; count++)
   cout << "Hello" << endl;
```

`for` Loop - Modifications

- You can declare variables in the *`initialization`* expression:

```
int sum = 0;
for (int num = 0; num <= 10;
num++)
      sum += num;
```

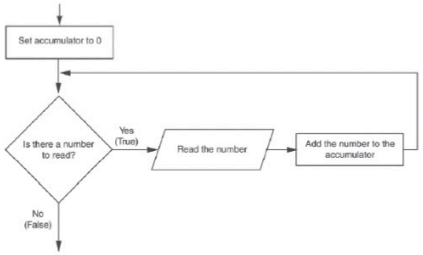
The scope of the variable `num` is the `for` loop.

Keeping a Running Total

- running total: accumulated sum of numbers from each repetition of loop
- accumulator: variable that holds running total

```
int sum=0, num=1; // sum is the
while (num <= 10) // accumulator
{   sum += num;
    num++;
}
cout << "Sum of numbers 1 - 10 is"
     << sum << endl;
```

Logic for Keeping a Running Total



Sentinels

- sentinel: value in a list of values that indicates end of data
- Special value that cannot be confused with a valid value, *e.g.*, `-999` for a test score
- Used to terminate input when user may not know how many values will be entered

A Sentinel in Program 5-13

Program 5-13

```
// This program calculates the total number of points a
// soccer team has earned over a series of games. The user
// enters a series of point values, then -1 when finished.
#include <iostream>
using namespace std;

int main()
{
   int game = 1,   // Game counter
       points,     // To hold a number of points
       total = 0;  // Accumulator

   cout << "Enter the number of points your team has earned\n";
   cout << "so far in the season, then enter -1 when finished.\n\n";
   cout << "Enter the points for game " << game << ": ";
   cin >> points;

   while (points != -1)
   {
      total += points;
      game++;
      cout << "Enter the points for game " << game << ": ";
      cin >> points;
   }
   cout << "\nThe total points are " << total << endl;
   return 0;
}
```

Continued…

A Sentinel in Program 5-13

Program Output with Example Input Shown in Bold

```
Enter the number of points your team has earned
so far in the season, then enter -1 when finished.

Enter the points for game 1: 7 [Enter]
Enter the points for game 2: 9 [Enter]
Enter the points for game 3: 4 [Enter]
Enter the points for game 4: 6 [Enter]
Enter the points for game 5: 8 [Enter]
Enter the points for game 6: -1 [Enter]

The total points are 34
```

Deciding Which Loop to Use

- The `while` loop is a conditional pretest loop
  - Iterates as long as a certain condition exits
  - Validating input
  - Reading lists of data terminated by a sentinel
- The `do-while` loop is a conditional posttest loop
  - Always iterates at least once
  - Repeating a menu
- The `for` loop is a pretest loop
  - Built-in expressions for initializing, testing, and updating
  - Situations where the exact number of iterations is known

Nested Loops - Notes

- Inner loop goes through all repetitions for each repetition of outer loop
- Inner loop repetitions complete sooner than outer loop
- Total number of repetitions for inner loop is product of number of repetitions of the two loops.

Using Files for Data Storage

- Can use files instead of keyboard, monitor screen for program input, output
- Allows data to be retained between program runs
- Steps:
  - *Open* the file
  - *Use* the file (read from, write to, or both)
  - *Close* the file

Files: What is Needed

- Use `fstream` header file for file access
- File stream types:
  - `ifstream` for input from a file
  - `ofstream` for output to a file
  - `fstream` for input from or output to a file
- Define file stream objects:

```
ifstream infile;
ofstream outfile;
```

Opening Files

- Create a link between file name (outside the program) and file stream object (inside the program)
- Use the `open` member function:

```
infile.open("inventory.dat");
outfile.open("report.txt");
```

- Filename may include drive, path info.
- Output file will be created if necessary; existing file will be erased first
- Input file must exist for `open` to work

Testing for File Open Errors

- Can test a file stream object to detect if an open operation failed:

```
infile.open("test.txt");
if (!infile)
{
  cout << "File open failure!";
}
```

- Can also use the `fail` member function

Using Files

- Can use output file object and << to send data to a file:

```
outfile << "Inventory report";
```

- Can use input file object and >> to copy data from file to variables:

```
infile >> partNum;
infile >> qtyInStock >>
qtyOnOrder;
```

Using Loops to Process Files

- The stream extraction operator >> returns `true` when a value was successfully read, `false` otherwise
- Can be tested in a `while` loop to continue execution as long as values are read from the file:

```
while (inputFile >> number) ...
```

Closing Files

- Use the `close` member function:

```
infile.close();
outfile.close();
```

- Don't wait for operating system to close files at program end:
  - may be limit on number of open files
  - may be buffered output data waiting to send to file

Letting the User Specify a Filename

- In many cases, you will want the user to specify the name of a file for the program to open.
- In C++ 11, you can pass a `string` object as an argument to a file stream object's `open` member function.

Letting the User Specify a Filename in Program 5-24

Program 5-24

```
1  // This program lets the user enter a filename.
2  #include <iostream>
3  #include <string>
4  #include <fstream>
5  using namespace std;
6
7  int main()
8  {
9      ifstream inputFile;
10     string filename;
11     int number;
12
13     // Get the filename from the user.
14     cout << "Enter the filename: ";
15     cin >> filename;
16
17     // Open the file.
18     inputFile.open(filename);
19
20     // If the file successfully opened, process it.
21     if (inputFile)
```

Continued...

Letting the User Specify a Filename in Program 5-24

```
22      {
23          // Read the numbers from the file and
24          // display them.
25          while (inputFile >> number)
26          {
27              cout << number << endl;
28          }
29
30          // Close the file.
31          inputFile.close();
32      }
33      else
34      {
35          // Display an error message.
36          cout << "Error opening the file.\n";
37      }
38      return 0;
39  }
```

Program Output with Example Input Shown in Bold

```
Enter the filename: ListOfNumbers.txt [Enter]
100
200
300
400
500
600
700
```

Using the `c_str` Member Function in Older Versions of C++

- Prior to C++ 11, the `open` member function requires that you pass the name of the file as a null-terminated string, which is also known as a *C-string*.
- *String literals are stored* in memory as null-terminated C-strings, but *string objects* are **not**.

Using the `c_str` Member Function in Older Versions of C++

- `string` objects have a member function named `c_str`
  - It returns the contents of the object formatted as a null-terminated C-string.
  - Here is the general format of how you call the `c_str` function:

```
stringObject.c_str()
```

  - Line 18 in Program 5-24 could be rewritten in the following manner:

```
inputFile.open(filename.c_str());
```

Breaking Out of a Loop

- Can use `break` to terminate execution of a loop
- Use sparingly if at all – makes code harder to understand and debug
- When used in an inner loop, terminates that loop only and goes back to outer loop

The `continue` Statement

- Can use `continue` to go to end of loop and prepare for next repetition
  - `while`, `do-while` loops: go to test, repeat loop if test passes
  - `for` loop: perform update step, then test, then repeat loop if test passes
- Use sparingly – like `break`, can make program logic hard to follow

Modular Programming

- Modular programming: breaking a program up into smaller, manageable functions or modules
- Function: a collection of statements to perform a task
- Motivation for modular programming:
  - Improves maintainability of programs
  - Simplifies the process of writing programs

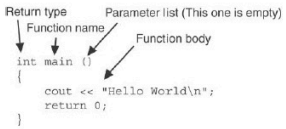
Defining and Calling Functions

- Function call: statement causes a function to execute
- Function definition: statements that make up a function

Function Definition

- Definition includes:
  - return type: data type of the value that function returns to the part of the program that called it
  - name: name of the function. Function names follow same rules as variables
  - parameter list: variables containing values passed to the function
  - body: statements that perform the function's task, enclosed in `{}`

Function Definition



Note: The line that reads `int main()` is the *function header*.

Function Return Type

- If a function returns a value, the type of the value must be indicated:

```
int main()
```

- If a function does not return a value, its return type is `void`:

```
void printHeading()
{
      cout << "Monthly Sales\n";
}
```

Calling a Function

- To call a function, use the function name followed by `()` and `;`

```
printHeading();
```

- When called, program executes the body of the called function
- After the function terminates, execution resumes in the calling function at point of call.

Functions in Program 6-1

Program 6-1

```
1  // This program has two functions: main and displayMessage
2  #include <iostream>
3  using namespace std;
4
5  //*****************************************
6  // Definition of function displayMessage  *
7  // This function displays a greeting.     *
8  //*****************************************
9
10 void displayMessage()
11 {
12    cout << "Hello from the function displayMessage.\n";
13 }
14
15 //*****************************************
16 // Function main                          *
17 //*****************************************
18
19 int main()
20 {
21    cout << "Hello from main.\n";
22    displayMessage();
23    cout << "Back in function main again.\n";
24    return 0;
25 }
```

Program Output

```
Hello from main.
Hello from the function displayMessage.
Back in function main again.
```

Calling Functions

- `main` can call any number of functions
- Functions can call other functions
- Compiler must know the following about a function before it is called:
  - name
  - return type
  - number of parameters
  - data type of each parameter

Function Prototypes

- Ways to notify the compiler about a function before a call to the function:
  - Place function definition before calling function's definition
  - Use a function prototype (function declaration) – like the function definition without the body
    - Header: `void printHeading()`
    - Prototype: `void printHeading();`

Prototype Notes

- Place prototypes near top of program
- Program must include either prototype or full function definition before any call to the function – compiler error otherwise
- When using prototypes, can place function definitions in any order in source file

Sending Data into a Function

- Can pass values into a function at time of call:

```
c = pow(a, b);
```

- Values passed to function are arguments
- Variables in a function that hold the values passed as arguments are parameters

A Function with a Parameter Variable

```
void displayValue(int num)
{
   cout << "The value is " << num << endl;
}
```

The integer variable `num` is a parameter.
It accepts any integer value passed to the function.

Function with a Parameter in Program 6-6

Program 6-6

```
1  // This program demonstrates a function with a parameter.
2  #include <iostream>
3  using namespace std;
4
5  // Function Prototype
6  void displayValue(int);
7
8  int main()
9  {
10    cout << "I am passing 5 to displayValue.\n";
11    displayValue(5);  // Call displayValue with argument 5
12    cout << "Now I am back in main.\n";
13    return 0;
14 }
15
```

(Program Continues)

Function with a Parameter in Program 6-6

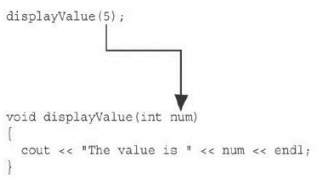
Program 6-6 *(continued)*

```
16 //*********************************************************
17 // Definition of function displayValue.                   *
18 // It uses an integer parameter whose value is displayed. *
19 //*********************************************************
20
21 void displayValue(int num)
22 {
23    cout << "The value is " << num << endl;
24 }
```

Program Output

```
I am passing 5 to displayValue.
The value is 5
Now I am back in main.
```

Function with a Parameter in Program 6-6



The function call in line 11 passes the value 5 as an argument to the function.

Other Parameter Terminology

- A parameter can also be called a formal parameter or a formal argument
- An argument can also be called an actual parameter or an actual argument

Parameters, Prototypes, and Function Headers

- For each function argument,
  - the prototype must include the data type of each parameter inside its parentheses
  - the header must include a declaration for each parameter in its `()`

```
void evenOrOdd(int);  //prototype
void evenOrOdd(int num) //header
evenOrOdd(val);       //call
```

Function Call Notes

- Value of argument is copied into parameter when the function is called
- A parameter's scope is the function which uses it
- Function can have multiple parameters
- There must be a data type listed in the prototype `()` and an argument declaration in the function header `()` for each parameter
- Arguments will be promoted/demoted as necessary to match parameters

Passing Multiple Arguments

When calling a function and passing multiple arguments:

- the number of arguments in the call must match the prototype and definition
- the first argument will be used to initialize the first parameter, the second argument to initialize the second parameter, etc.

Passing Multiple Arguments in Program 6-8

Program 6-8

```
1  // This program demonstrates a function with three parameters.
2  #include <iostream>
3  using namespace std;
4
5  // Function Prototype
6  void showSum(int, int, int);
7
8  int main()
9  {
10    int value1, value2, value3;
11
12    // Get three integers.
13    cout << "Enter three integers and I will display ";
14    cout << "their sum: ";
15    cin >> value1 >> value2 >> value3;
16
17    // Call showSum passing three arguments.
18    showSum(value1, value2, value3);
19    return 0;
20 }
21
```

(Program Continues)

Passing Multiple Arguments in Program 6-8

```
22 //*********************************************************
23 // Definition of function showSum.                        *
24 // It uses three integer parameters. Their sum is displayed. *
25 //*********************************************************
26
27 void showSum(int num1, int num2, int num3)
28 {
29    cout << (num1 + num2 + num3) << endl;
30 }
```

Program Output with Example Input Shown in Bold

```
Enter three integers and I will display their sum: 4 8 7 [Enter]
19
```

Passing Data by Value

- Pass by value: when an argument is passed to a function, its value is copied into the parameter.
- Changes to the parameter in the function do not affect the value of the argument

Passing Information to Parameters by Value

- Example: `int val=5;`
  `evenOrOdd(val);`

- `evenOrOdd` can change variable `num`, but it will have no effect on variable `val`

Using Functions in Menu-Driven Programs

- Functions can be used
  - to implement user choices from menu
  - to implement general-purpose tasks:
    - Higher-level functions can call general-purpose functions, minimizing the total number of functions and speeding program development time
- *See Program 6-10 in the book*

The `return` Statement

- Used to end execution of a function
- Can be placed anywhere in a function
  - Statements that follow the `return` statement will not be executed
- Can be used to prevent abnormal termination of program
- In a `void` function without a `return` statement, the function ends at its last }

Returning a Value From a Function

- A function can return a value back to the statement that called the function.
- You've already seen the `pow` function, which returns a value:

```
double x;
x = pow(2.0, 10.0);
```

Returning a Value From a Function

- In a value-returning function, the `return` statement can be used to return a value from function to the point of call. Example:

```
int sum(int num1, int num2)
{
  double result;
  result = num1 + num2;
  return result;
}
```

A Value-Returning Function

Return Type

```
int sum(int num1, int num2)
{
   double result;
   result = num1 + num2;
   return result;
}
```

Value Being Returned

A Value-Returning Function

```
int sum(int num1, int num2)
{
   return num1 + num2;
}
```

Functions can return the values of expressions, such as `num1 + num2`

Returning a Value From a Function

- The prototype and the definition must indicate the data type of return value (not `void`)
- Calling function should use return value:
  - assign it to a variable
  - send it to `cout`
  - use it in an expression

Returning a Boolean Value

- Function can return `true` or `false`
- Declare return type in function prototype and heading as `bool`
- Function body must contain `return` statement(s) that return `true` or `false`
- Calling function can use return value in a relational expression

Returning a Boolean Value in Program 6-15

Program 6-15

```
1  // This program uses a function that returns true or false.
2  #include <iostream>
3  using namespace std;
4
5  // Function prototype
6  bool isEven(int);
7
8  int main()
9  {
10    int val;
11
12    // Get a number from the user.
13    cout << "Enter an integer and I will tell you ";
14    cout << "if it is even or odd: ";
15    cin >> val;
16
17    // Indicate whether it is even or odd.
18    if (isEven(val))
19       cout << val << " is even.\n";
20    else
21       cout << val << " is odd.\n";
22    return 0;
23 }
24
```

(Program Continues)

Returning a Boolean Value in Program 6-15

```
25 //*************************************************************
26 // Definition of function isEven. This function accepts an    *
27 // integer argument and tests it to be even or odd. The function *
28 // returns true if the argument is even or false if the argument *
29 // is odd. The return value is a bool.                        *
30 //*************************************************************
31
32 bool isEven(int number)
33 {
34    bool status;
35
36    if (number % 2 == 0)
37       status = true;  // The number is even if there is no remainder.
38    else
39       status = false; // Otherwise, the number is odd.
40    return status;
41 }
```

Program Output with Example Input Shown in Bold

Enter an integer and I will tell you if it is even or odd: **5 [Enter]**
5 is odd.

Local and Global Variables

- Variables defined inside a function are *local* to that function. They are hidden from the statements in other functions, which normally cannot access them.
- Because the variables defined in a function are hidden, other functions may have separate, distinct variables with the same name.

Local Variables in Program 6-16

Program 6-16

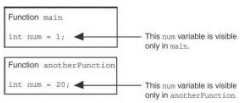
```
1  // This program shows that variables defined in a function
2  // are hidden from other functions.
3  #include <iostream>
4  using namespace std;
5
6  void anotherFunction(); // Function prototype
7
8  int main()
9  {
10    int num = 1;  // Local variable
11
12    cout << "In main, num is " << num << endl;
13    anotherFunction();
14    cout << "Back in main, num is " << num << endl;
15    return 0;
16 }
17
18 //*************************************************
19 // Definition of anotherFunction                  *
20 // It has a local variable, num, whose initial value *
21 // is displayed.                                  *
22 //*************************************************
23
24 void anotherFunction()
25 {
26    int num = 20;  // Local variable
27
28    cout << "In anotherFunction, num is " << num << endl;
29 }
```

Local Variables in Program 6-16

Program Output

```
In main, num is 1
In anotherFunction, num is 20
Back in main, num is 1
```

When the program is executing in `main`, the `num` variable defined in `main` is visible. When `anotherFunction` is called, however, only variables defined inside it are visible, so the `num` variable in `main` is hidden.



Local Variable Lifetime

- A function's local variables exist only while the function is executing. This is known as the *lifetime* of a local variable.
- When the function begins, its local variables and its parameter variables are created in memory, and when the function ends, the local variables and parameter variables are destroyed.
- This means that any value stored in a local variable is lost between calls to the function in which the variable is declared.

Global Variables and Global Constants

- A global variable is any variable defined outside all the functions in a program.
- The scope of a global variable is the portion of the program from the variable definition to the end.
- This means that a global variable can be accessed by *all* functions that are defined after the global variable is defined.

Global Variables and Global Constants

- You should avoid using global variables because they make programs difficult to debug.
- Any global that you create should be *global constants*.

Global Constants in Program 6-19

Program 6-19

Global constants defined for values that do not change throughout the program's execution.

```
1  // This program calculates gross pay.
2  #include <iostream>
3  #include <iomanip>
4  using namespace std;
5
6  // Global constants
7  const double PAY_RATE = 22.55;    // Hourly pay rate
8  const double BASE_HOURS = 40.0;   // Max non-overtime hours
9  const double OT_MULTIPLIER = 1.5; // Overtime multiplier
10
11 // Function prototypes
12 double getBasePay(double);
13 double getOvertimePay(double);
14
15 int main()
16 {
17    double hours,           // Hours worked
18           basePay,         // Base pay
19           overtime = 0.0,  // Overtime pay
20           totalPay;        // Total pay
```

Global Constants in Program 6-19

The constants are then used for those values throughout the program.

```
29    // Get overtime pay, if any.
30    if (hours > BASE_HOURS)
31       overtime = getOvertimePay(hours);
```

```
56 // Determine base pay.
57 if (hoursWorked > BASE_HOURS)
58    basePay = BASE_HOURS * PAY_RATE;
59 else
60    basePay = hoursWorked * PAY_RATE;
```

```
75    // Determine overtime pay.
76    if (hoursWorked > BASE_HOURS)
77    {
78       overtimePay = (hoursWorked - BASE_HOURS) *
79                     PAY_RATE * OT_MULTIPLIER;
```

Initializing Local and Global Variables

- Local variables are not automatically initialized. They must be initialized by programmer.
- Global variables (not constants) are automatically initialized to `0` (numeric) or `NULL` (character) when the variable is defined.

Static Local Variables

- Local variables only exist while the function is executing. When the function terminates, the contents of local variables are lost.
- `static` local variables retain their contents between function calls.
- `static` local variables are defined and initialized only the first time the function is executed. `0` is the default initialization value.

Local Variables Do Not Retain Values Between Function calls in Program 6-21

Program 6-21

```
// This program shows that local variables do not retain
// their values between function calls.
#include <iostream>
using namespace std;

// Function prototype
void showLocal();

int main()
{
   showLocal();
   showLocal();
   return 0;
}
```

(Program Continues)

Local Variables Do Not Retain Values Between Function calls in Program 6-21

Program 6-21 *(continued)*

```
//*************************************************************
// Definition of function showLocal.                          *
// The initial value of localNum, which is 5, is displayed.   *
// The value of localNum is then changed to 99 before the     *
// function returns.                                          *
//*************************************************************

void showLocal()
{
   int localNum = 5; // Local variable

   cout << "localNum is " << localNum << endl;
   localNum = 99;
}
```

Program Output

```
localNum is 5
localNum is 5
```

In this program, each time `showLocal` is called, the `localNum` variable is re-created and initialized with the value 5.

A Different Approach, Using a Static Variable in Program 6-22

Program 6-22

```
// This program uses a static local variable.
#include <iostream>
using namespace std;

void showStatic(); // Function prototype

int main()
{
   // Call the showStatic function five times.
   for (int count = 0; count < 5; count++)
      showStatic();
   return 0;
}
```

(Program Continues)

A Different Approach, Using a Static Variable in Program 6-22

Program 6-22 *(continued)*

```
//*************************************************************
// Definition of function showStatic.                         *
// statNum is a static local variable. Its value is displayed *
// and then incremented just before the function returns.     *
//*************************************************************

void showStatic()
{
   static int statNum;

   cout << "statNum is " << statNum << endl;
   statNum++;
}
```

Program Output

```
statNum is 0
statNum is 1
statNum is 2
statNum is 3
statNum is 4
```

`statNum` is automatically initialized to 0. Notice that it retains its value between function calls.

If you do initialize a local static variable, the initialization only happens once. See Program 6-23.

```
//*************************************************************
// Definition of function showStatic.                         *
// statNum is a static local variable. Its value is displayed *
// and then incremented just before the function returns.     *
//*************************************************************

void showStatic()
{
   static int statNum = 5;

   cout << "statNum is " << statNum << endl;
   statNum++;
}
```

Program Output

```
statNum is 5
statNum is 6
statNum is 7
statNum is 8
statNum is 9
```

Default Arguments

A Default argument is an argument that is passed automatically to a parameter if the argument is missing on the function call.

- Must be a constant declared in prototype:
  `void evenOrOdd(int = 0);`
- Can be declared in header if no prototype
- Multi-parameter functions may have default arguments for some or all of them:
  `int getSum(int, int=0, int=0);`

Default Arguments in Program 6-24

Default arguments specified in the prototype

Program 6-24

```
// This program demonstrates default function arguments.
#include <iostream>
using namespace std;

// Function prototype with default arguments
void displayStars(int = 10, int = 1);

int main()
{
   displayStars();      // Use default values for cols and rows.
   cout << endl;
   displayStars(5);     // Use default value for rows.
   cout << endl;
   displayStars(7, 3);  // Use 7 for cols and 3 for rows.
   return 0;
}
```

(Program Continues)

Default Arguments in Program 6-24

```
//*************************************************************
// Definition of function displayStars.                       *
// The default argument for cols is 10 and for rows is 1.     *
// This function displays a square made of asterisks.         *
//*************************************************************

void displayStars(int cols, int rows)
{
   // Nested loop. The outer loop controls the rows
   // and the inner loop controls the columns.
   for (int down = 0; down < rows; down++)
   {
      for (int across = 0; across < cols; across++)
         cout << "*";
      cout << endl;
   }
}
```

Program Output

```
**********

*****

*******
*******
*******
```

Default Arguments

- If not all parameters to a function have default values, the defaultless ones are declared first in the parameter list:

```
int getSum(int, int=0, int=0);// OK
int getSum(int, int=0, int);  // NO
```

- When an argument is omitted from a function call, all arguments after it must also be omitted:

```
sum = getSum(num1, num2);     // OK
sum = getSum(num1, , num3);   // NO
```

Using Reference Variables as Parameters

- A mechanism that allows a function to work with the original argument from the function call, not a copy of the argument
- Allows the function to modify values stored in the calling environment
- Provides a way for the function to 'return' more than one value

Passing by Reference

- A reference variable is an alias for another variable
- Defined with an ampersand (&)

```
void getDimensions(int&, int&);
```

- Changes to a reference variable are made to the variable it refers to
- Use reference variables to implement passing parameters *by reference*

Passing a Variable By Reference in Program 6-25

The & here in the prototype indicates that the parameter is a reference variable.

Program 6-25

```
1  // This program uses a reference variable as a function
2  // parameter.
3  #include <iostream>
4  using namespace std;
5
6  // Function prototype. The parameter is a reference variable.
7  void doubleNum(int &);
8
9  int main()
10 {
11    int value = 4;
12
13    cout << "In main, value is " << value << endl;
14    cout << "Now calling doubleNum..." << endl;
15    doubleNum(value);
16    cout << "Now back in main. value is " << value << endl;
17    return 0;
18 }
19
```

Here we are passing value by reference.

(Program Continues)

Passing a Variable By Reference in Program 6-25

The & also appears here in the function header.

```
20 //*************************************************************
21 // Definition of doubleNum.                                   *
22 // The parameter refVar is a reference variable. The value    *
23 // in refVar is doubled.                                      *
24 //*************************************************************
25
26 void doubleNum (int &refVar)
27 {
28    refVar *= 2;
29 }
```

Program Output

```
In main, value is 4
Now calling doubleNum...
Now back in main. value is 8
```

Reference Variable Notes

- Each reference parameter must contain &
- Space between type and & is unimportant
- Must use & in both prototype and header
- Argument passed to reference parameter must be a variable – cannot be an expression or constant
- Use when appropriate – don't use when argument should not be changed by function, or if function needs to return only 1 value

Overloading Functions

- Overloaded functions have the same name but different parameter lists
- Can be used to create functions that perform the same task but take different parameter types or different number of parameters
- Compiler will determine which version of function to call by argument and parameter lists

Function Overloading Examples

Using these overloaded functions,

```
void getDimensions(int);              // 1
void getDimensions(int, int);         // 2
void getDimensions(int, double);      // 3
void getDimensions(double, double);// 4
```

the compiler will use them as follows:

```
int length, width;
double base, height;
getDimensions(length);              // 1
getDimensions(length, width);       // 2
getDimensions(length, height);      // 3
getDimensions(height, base);        // 4
```

Function Overloading in Program 6-27

Program 6-27

```
1  // This program uses overloaded functions.
2  #include <iostream>
3  #include <iomanip>
4  using namespace std;
5
6  // Function prototypes
7  int square(int);
8  double square(double);
9
10 int main()
11 {
12    int userInt;
13    double userFloat;
14
15    // Get an int and a double.
16    cout << fixed << showpoint << setprecision(2);
17    cout << "Enter an integer and a floating-point value: ";
18    cin >> userInt >> userFloat;
19
20    // Display their squares.
21    cout << "Here are their squares: ";
22    cout << square(userInt) << " and " << square(userFloat);
23    return 0;
24 }
```

The overloaded functions have different parameter lists

Passing a **double**

Passing an **int**

(Program Continues)

Function Overloading in Program 6-27

```
26 //*************************************************************
27 // Definition of overloaded function square.                  *
28 // This function uses an int parameter, number. It returns the *
29 // square of number as an int.                                *
30 //*************************************************************
31
32 int square(int number)
33 {
34    return number * number;
35 }
36
37 //*************************************************************
38 // Definition of overloaded function square.                  *
39 // This function uses a double parameter, number. It returns  *
40 // the square of number as a double.                          *
41 //*************************************************************
42
43 double square(double number)
44 {
45    return number * number;
46 }
```

Program Output with Example Input Shown in Bold

Enter an integer and a floating-point value: **12 4.2 [Enter]**
Here are their squares: 144 and 17.64

The `exit()` Function

- Terminates the execution of a program
- Can be called from any function
- Can pass an `int` value to operating system to indicate status of program termination
- Usually used for abnormal termination of program
- Requires `cstdlib` header file

The `exit()` Function

- Example:

```
exit(0);
```

- The `cstdlib` header defines two constants that are commonly passed, to indicate success or failure:

```
exit(EXIT_SUCCESS);
exit(EXIT_FAILURE);
```

Stubs and Drivers

- Useful for testing and debugging program and function logic and design
- Stub: A dummy function used in place of an actual function
  - Usually displays a message indicating it was called. May also display parameters
- Driver: A function that tests another function by calling it
  - Various arguments are passed and return values are tested

Arrays Hold Multiple Values

- Array: variable that can store multiple values of the same type
- Values are stored in adjacent memory locations
- Declared using `[]` operator:

```
int tests[5];
```

Array Terminology

In the definition `int tests[5];`

- `int` is the data type of the array elements
- `tests` is the name of the array
- `5`, in `[5]`, is the size declarator. It shows the number of elements in the array.
- The size of an array is (number of elements) * (size of each element)

Array Terminology

- The size of an array is:
  - the total number of bytes allocated for it
  - (number of elements) * (number of bytes for each element)
- Examples:

`int tests[5]` is an array of 20 bytes, assuming 4 bytes for an `int`

`long double measures[10]` is an array of 80 bytes, assuming 8 bytes for a `long double`

Size Declarators

- Named constants are commonly used as size declarators.

```
const int SIZE = 5;
int tests[SIZE];
```

- This eases program maintenance when the size of the array needs to be changed.

Accessing Array Contents

- Can access element with a constant or literal subscript:

```
cout << tests[3] << endl;
```

- Can use integer expression as subscript:

```
int i = 5;
cout << tests[i] << endl;
```

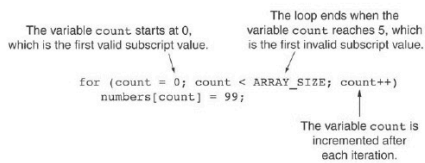
Using a Loop to Step Through an Array

- Example – The following code defines an array, `numbers`, and assigns 99 to each element:

```
const int ARRAY_SIZE = 5;
int numbers[ARRAY_SIZE];

for (int count = 0; count < ARRAY_SIZE; count++)
     numbers[count] = 99;
```

A Closer Look At the Loop



Default Initialization

- Global array → all elements initialized to `0` by default
- Local array → all elements *uninitialized* by default

Array Initialization

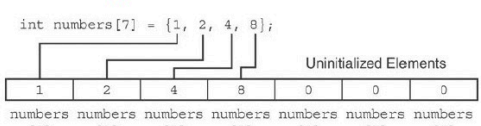
- Arrays can be initialized with an initialization list:

```
const int SIZE = 5;
int tests[SIZE] = {79,82,91,77,84};
```

- The values are stored in the array in the order in which they appear in the list.
- The initialization list cannot exceed the array size.

Partial Array Initialization

- If array is initialized with fewer initial values than the size declarator, the remaining elements will be set to `0`:



Implicit Array Sizing

- Can determine array size by the size of the initialization list:

```
int quizzes[]={12,17,15,11};
```

12	17	15	11

- Must use either array size declarator or initialization list at array definition

No Bounds Checking in C++

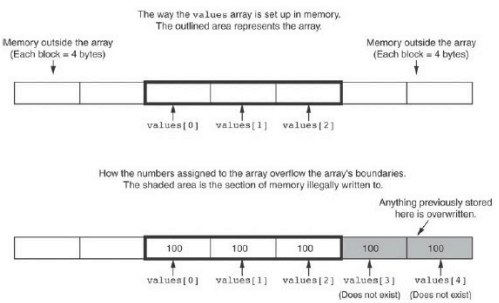
- When you use a value as an array subscript, C++ does not check it to make sure it is a *valid* subscript.
- In other words, you can use subscripts that are beyond the bounds of the array.

Code From Program 7-9

- The following code defines a three-element array, and then writes five values to it!

```
9       const int SIZE = 3;   // Constant for the array size
10      int values[SIZE];     // An array of 3 integers
11      int count;            // Loop counter variable
12
13      // Attempt to store five numbers in the 3-element array.
14      cout << "I will store 5 numbers in a 3-element array!\n";
15      for (count = 0; count < 5; count++)
16          values[count] = 100;
```

What the Code Does



No Bounds Checking in C++

- Be careful not to use invalid subscripts.
- Doing so can corrupt other memory locations, crash program, or lock up computer, and cause elusive bugs.

Off-By-One Errors

- An off-by-one error happens when you use array subscripts that are off by one.
- This can happen when you start subscripts at 1 rather than 0:

```
// This code has an off-by-one error.
const int SIZE = 100;
int numbers[SIZE];
for (int count = 1; count <= SIZE; count++)
   numbers[count] = 0;
```

The Range-Based `for` Loop

- C++ 11 provides a specialized version of the `for` loop that, in many circumstances, simplifies array processing.
- *The range-based `for` loop is a loop that iterates once for each element in an array.*
- *Each time the loop iterates, it copies an element from the array to a built-in variable, known as the range variable.*
- The range-based `for` loop automatically knows the number of elements in an array.
  - You do not have to use a counter variable.
  - You do not have to worry about stepping outside the bounds of the array.

The Range-Based `for` Loop

- Here is the general format of the range-based for loop:

```
for (dataType rangeVariable : array)
      statement;
```

- ***dataType*** is the data type of the range variable.
- ***rangeVariable*** is the name of the range variable. This variable will receive the value of a different array element during each loop iteration.
- ***array*** is the name of an array on which you wish the loop to operate.
- ***statement*** is a statement that executes during a loop iteration. If you need to execute more than one statement in the loop, enclose the statements in a set of braces.

The range-based `for` loop in Program 7-10

```
1   // This program demonstrates the range-based for loop.
2   #include <iostream>
3   using namespace std;
4
5   int main()
6   {
7       // Define an array of integers.
8       int numbers[] = { 10, 20, 30, 40, 50 };
9
10      // Display the values in the array.
11      for (int val : numbers)
12          cout << val << endl;
13
14      return 0;
15  }
```

Modifying an Array with a Range-Based `for` Loop

- As the range-based `for` loop executes, its range variable contains only a copy of an array element.
- You cannot use a range-based `for` loop to modify the contents of an array unless you declare the range variable as a reference.
- To declare the range variable as a reference variable, simply write an ampersand (`&`) in front of its name in the loop header.
- Program 7-12 demonstrates

Modifying an Array with a Range-Based `for` Loop in Program 7-12

```
const int SIZE = 5;
int numbers[5];

// Get values for the array.
for (int &val : numbers)
{
   cout << "Enter an integer value: ";
   cin >> val;
}

// Display the values in the array.
cout << "Here are the values you entered:\n";
for (int val : numbers)
   cout << val << endl;
```

Modifying an Array with a Range-Based `for` Loop

You can use the `auto` key word with a reference range variable. For example, the code in lines 12 through 16 in Program 7-12 could have been written like this:

```
for (auto &val : numbers)
{
   cout << "Enter an integer value: ";
   cin >> val;
}
```

The Range-Based `for` Loop versus the Regular `for` Loop

- The range-based for loop can be used in any situation where you need to step through the elements of an array, and you do not need to use the element subscripts.
- If you need the element subscript for some purpose, use the regular `for` loop.

Processing Array Contents

- Array elements can be treated as ordinary variables of the same type as the array
- When using ++, -- operators, don't confuse the element with the subscript:

```
tests[i]++; // add 1 to tests[i]
tests[i++]; // increment i, no
            // effect on tests
```

Array Assignment

To copy one array to another,

- Don't try to assign one array to the other:

```
newTests = tests;  // Won't work
```

- Instead, assign element-by-element:

```
for (i = 0; i < ARRAY_SIZE; i++)
   newTests[i] = tests[i];
```

Printing the Contents of an Array

- You can display the contents of a *character* array by sending its name to cout:

```
char fName[] = "Henry";
cout << fName << endl;
```

But, this ONLY works with character arrays!

Printing the Contents of an Array

- For other types of arrays, you must print element-by-element:

```
for (i = 0; i < ARRAY_SIZE; i++)
   cout << tests[i] << endl;
```

Printing the Contents of an Array

- In C++ 11 you can use the range-based `for` loop to display an array's contents, as shown here:

```
for (int val : numbers)
   cout << val << endl;
```

Summing and Averaging Array Elements

- Use a simple loop to add together array elements:

```
int tnum;
double average, sum = 0;
for(tnum = 0; tnum < SIZE; tnum++)
      sum += tests[tnum];
```

- Once summed, can compute average:

```
average = sum / SIZE;
```

Summing and Averaging Array Elements

- In C++ 11 you can use the range-based `for` loop, as shown here:

```
double total = 0;  // Initialize accumulator
double average;    // Will hold the average
for (int val : scores)
    total += val;
average = total / NUM_SCORES;
```

Finding the Highest Value in an Array

```
int count;
int highest;
highest = numbers[0];
for (count = 1; count < SIZE; count++)
{
   if (numbers[count] > highest)
      highest = numbers[count];
}
```

When this code is finished, the `highest` variable will contains the highest value in the `numbers` array.

Finding the Lowest Value in an Array

```
int count;
int lowest;
lowest = numbers[0];
for (count = 1; count < SIZE; count++)
{
   if (numbers[count] < lowest)
      lowest = numbers[count];
}
```

When this code is finished, the `lowest` variable will contains the lowest value in the `numbers` array.

Partially-Filled Arrays

- If it is unknown how much data an array will be holding:
  - Make the array large enough to hold the largest expected number of elements.
  - Use a counter variable to keep track of the number of items stored in the array.

Comparing Arrays

- To compare two arrays, you must compare element-by-element:

```
const int SIZE = 5;
int firstArray[SIZE] = { 5, 10, 15, 20, 25 };
int secondArray[SIZE] = { 5, 10, 15, 20, 25 };
bool arraysEqual = true; // Flag variable
int count = 0;           // Loop counter variable
// Compare the two arrays.
while (arraysEqual && count < SIZE)
{
   if (firstArray[count] != secondArray[count])
      arraysEqual = false;
   count++;
}
if (arraysEqual)
   cout << "The arrays are equal.\n";
else
   cout << "The arrays are not equal.\n";
```

Using Parallel Arrays

- Parallel arrays: two or more arrays that contain related data
- A subscript is used to relate arrays: elements at same subscript are related
- Arrays may be of different types

Parallel Array Example

```
const int SIZE = 5;    // Array size
int id[SIZE];          // student ID
double average[SIZE]; // course average
char grade[SIZE];      // course grade
...
for(int i = 0; i < SIZE; i++)
{
   cout << "Student ID: " << id[i]
        << " average: " << average[i]
        << " grade: " << grade[i]
        << endl;
}
```

Parallel Arrays in Program 7-15

Program 7-15

```
// This program uses two parallel arrays: one for hours
// worked and one for pay rate.
#include <iostream>
#include <iomanip>
using namespace std;

int main()
{
    const int NUM_EMPLOYEES = 5;  // Number of employees
    int hours[NUM_EMPLOYEES];     // Holds hours worked
    double payRate[NUM_EMPLOYEES]; // Holds pay rates

    // Input the hours worked and the hourly pay rate.
    cout << "Enter the hours worked by " << NUM_EMPLOYEES
         << " employees and their\n"
         << "hourly pay rates.\n";
    for (int index = 0; index < NUM_EMPLOYEES; index++)
    {
        cout << "Hours worked by employee #" << (index+1) << ": ";
        cin >> hours[index];
        cout << "Hourly pay rate for employee #" << (index+1) << ": ";
        cin >> payRate[index];
    }
```

(Program Continues)

Parallel Arrays in Program 7-15

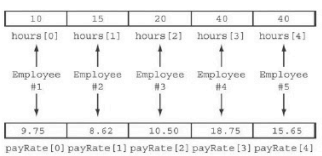
```
    // Display each employee's gross pay.
    cout << "Here is the gross pay for each employee:\n";
    cout << fixed << showpoint << setprecision(2);
    for (int index = 0; index < NUM_EMPLOYEES; index++)
    {
        double grossPay = hours[index] * payRate[index];
        cout << "Employee #" << (index + 1);
        cout << ": $" << grossPay << endl;
    }
    return 0;
}
```

Program Output with Example Input Shown in Bold

```
Enter the hours worked by 5 employees and their
hourly pay rates.
Hours worked by employee #1: 10 [Enter]
Hourly pay rate for employee #1: 9.75 [Enter]
Hours worked by employee #2: 15 [Enter]
Hourly pay rate for employee #2: 8.62 [Enter]
Hours worked by employee #3: 20 [Enter]
Hourly pay rate for employee #3: 10.50 [Enter]
Hours worked by employee #4: 40 [Enter]
Hourly pay rate for employee #4: 18.75 [Enter]
Hours worked by employee #5: 40 [Enter]
Hourly pay rate for employee #5: 15.65 [Enter]
Here is the gross pay for each employee:
Employee #1: $97.50
Employee #2: $129.30
Employee #3: $210.00
Employee #4: $750.00
Employee #5: $626.00
```

Parallel Arrays in Program 7-15

The `hours` and `payRate` arrays are related through their subscripts:



Arrays as Function Arguments

- To pass an array to a function, just use the array name:

```
showScores(tests);
```

- To define a function that takes an array parameter, use empty `[]` for array argument:

```
// function prototype
void showScores(int []);

// function header
void showScores(int tests[])
```

Arrays as Function Arguments

- When passing an array to a function, it is common to pass array size so that function knows how many elements to process:

```
showScores(tests, ARRAY_SIZE);
```

- Array size must also be reflected in prototype, header:

```
// function prototype
void showScores(int [], int);

// function header
void showScores(int tests[], int size)
```


Passing an Array to a Function in Program 7-17

Program 7-17

```
// This program demonstrates an array being passed to a function.
#include <iostream>
using namespace std;

void showValues(int [], int); // Function prototype

int main()
{
    const int ARRAY_SIZE = 8;
    int numbers[ARRAY_SIZE] = {5, 10, 15, 20, 25, 30, 35, 40};

    showValues(numbers, ARRAY_SIZE);
    return 0;
}

//*********************************************************
// Definition of function showValues.                     *
// This function accepts an array of integers and         *
// the array's size as its arguments. The contents        *
// of the array are displayed.                            *
//*********************************************************

void showValues(int nums[], int size)
{
    for (int index = 0; index < size; index++)
        cout << nums[index] << " ";
    cout << endl;
}
```

Program Output

```
5 10 15 20 25 30 35 40
```

Modifying Arrays in Functions

- Array names in functions are like reference variables – changes made to array in a function are reflected in actual array in calling function
- Need to exercise caution that array is not inadvertently changed by a function

Two-Dimensional Arrays

- Can define one array for multiple sets of data
- Like a table in a spreadsheet
- Use two size declarators in definition:

```
const int ROWS = 4, COLS = 3;
int exams[ROWS][COLS];
```

- First declarator is number of rows; second is number of columns

Two-Dimensional Array Representation

```
const int ROWS = 4, COLS = 3; int exams[ROWS][COLS];
```

rows \ columns			
	exams[0][0]	exams[0][1]	exams[0][2]
	exams[1][0]	exams[1][1]	exams[1][2]
	exams[2][0]	exams[2][1]	exams[2][2]
	exams[3][0]	exams[3][1]	exams[3][2]

- Use two subscripts to access element:

```
exams[2][2] = 86;
```

2D Array Initialization

- Two-dimensional arrays are initialized row-by-row:

```
const int ROWS = 2, COLS = 2;
int exams[ROWS][COLS] = { {84, 78},
                          {92, 97} };
```

84	78
92	97

- Can omit inner `{ }`, some initial values in a row – array elements without initial values will be set to `0` or `NULL`

Two-Dimensional Array as Parameter, Argument

- Use array name as argument in function call:

```
getExams(exams, 2);
```

- Use empty `[]` for row, size declarator for column in prototype, header:

```
const int COLS = 2;
// Prototype
void getExams(int [][COLS], int);

// Header
void getExams(int exams[][COLS], int rows)
```

Example – The `showArray` Function from Program 7-22

```
30  //*****************************************************************
31  // Function Definition for showArray                              *
32  // The first argument is a two-dimensional int array with COLS    *
33  // columns. The second argument, rows, specifies the number of    *
34  // rows in the array. The function displays the array's contents. *
35  //*****************************************************************
36
37  void showArray(int array[][COLS], int rows)
38  {
39     for (int x = 0; x < rows; x++)
40     {
41        for (int y = 0; y < COLS; y++)
42        {
43           cout << setw(4) << array[x][y] << " ";
44        }
45        cout << endl;
46     }
47  }
```

How `showArray` is Called

```
15     int table1[TBL1_ROWS][COLS] = {{1, 2, 3, 4},
16                                    {5, 6, 7, 8},
17                                    {9, 10, 11, 12}};
18     int table2[TBL2_ROWS][COLS] = {{10, 20, 30, 40},
19                                    {50, 60, 70, 80},
20                                    {90, 100, 110, 120},
21                                    {130, 140, 150, 160}};
22
23     cout << "The contents of table1 are:\n";
24     showArray(table1, TBL1_ROWS);
25     cout << "The contents of table2 are:\n";
26     showArray(table2, TBL2_ROWS);
```

Summing All the Elements in a Two-Dimensional Array

- Given the following definitions:

```
const int NUM_ROWS = 5; // Number of rows
const int NUM_COLS = 5; // Number of columns
int total = 0;          // Accumulator
int numbers[NUM_ROWS][NUM_COLS] =
   {{2, 7, 9, 6, 4},
    {6, 1, 8, 9, 4},
    {4, 3, 7, 2, 9},
    {9, 9, 0, 3, 1},
    {6, 2, 7, 4, 1}};
```

Summing All the Elements in a Two-Dimensional Array

```
// Sum the array elements.
for (int row = 0; row < NUM_ROWS; row++)
{
   for (int col = 0; col < NUM_COLS; col++)
      total += numbers[row][col];
}

// Display the sum.
cout << "The total is " << total << endl;
```

Summing the Rows of a Two-Dimensional Array

- Given the following definitions:

```
const int NUM_STUDENTS = 3;
const int NUM_SCORES = 5;
double total;   // Accumulator
double average; // To hold average scores
double scores[NUM_STUDENTS][NUM_SCORES] =
     {{88, 97, 79, 86, 94},
      {86, 91, 78, 79, 84},
      {82, 73, 77, 82, 89}};
```

Summing the Rows of a Two-Dimensional Array

```
// Get each student's average score.
for (int row = 0; row < NUM_STUDENTS; row++)
{
   // Set the accumulator.
   total = 0;
   // Sum a row.
   for (int col = 0; col < NUM_SCORES; col++)
      total += scores[row][col];
   // Get the average
   average = total / NUM_SCORES;
   // Display the average.
   cout << "Score average for student "
        << (row + 1) << " is " << average <<endl;
}
```

Summing the Columns of a Two-Dimensional Array

- Given the following definitions:

```
const int NUM_STUDENTS = 3;
const int NUM_SCORES = 5;
double total;   // Accumulator
double average; // To hold average scores
double scores[NUM_STUDENTS][NUM_SCORES] =
     {{88, 97, 79, 86, 94},
      {86, 91, 78, 79, 84},
      {82, 73, 77, 82, 89}};
```

Summing the Columns of a Two-Dimensional Array

```
// Get the class average for each score.
for (int col = 0; col < NUM_SCORES; col++)
{
   // Reset the accumulator.
   total = 0;
   // Sum a column
   for (int row = 0; row < NUM_STUDENTS; row++)
      total += scores[row][col];
   // Get the average
   average = total / NUM_STUDENTS;
   // Display the class average.
   cout << "Class average for test " << (col + 1)
        << " is " << average << endl;
}
```

Arrays with Three or More Dimensions

- Can define arrays with any number of dimensions:

```
short rectSolid[2][3][5];
double timeGrid[3][4][3][4];
```

- When used as parameter, specify all but 1st dimension in prototype, heading:

```
void getRectSolid(short [][3][5]);
```

Introduction to the STL `vector`

- A data type defined in the Standard Template Library (covered more in Chapter 17)
- Can hold values of any type:

```
vector<int> scores;
```

- Automatically adds space as more is needed – no need to determine size at definition
- Can use `[]` to access elements

Declaring Vectors

- You must `#include<vector>`
- Declare a vector to hold `int` element:

```
vector<int> scores;
```

- Declare a vector with initial size 30:

```
vector<int> scores(30);
```

- Declare a vector and initialize all elements to 0:

```
vector<int> scores(30, 0);
```

- Declare a vector initialized to size and contents of another vector:

```
vector<int> finals(scores);
```

Adding Elements to a Vector

- If you are using C++ 11, you can initialize a vector with a list of values:

```
vector<int> numbers { 10, 20, 30, 40 };
```

- Use `push_back` member function to add element to a full array or to an array that had no defined size:

```
scores.push_back(75);
```

- Use `size` member function to determine size of a vector:

```
howbig = scores.size();
```

Removing Vector Elements

- Use `pop_back` member function to remove last element from vector:

```
scores.pop_back();
```

- To remove all contents of vector, use `clear` member function:

```
scores.clear();
```

- To determine if vector is empty, use `empty` member function:

```
while (!scores.empty()) ...
```

Using the Range-Based `for` Loop with a vector

Program 7-25

```
// This program demonstrates the range-based for loop with a vector.
#include <iostream>
#include <vector>
using namespace std;

int main()
{
    // Define and initialize a vector.
    vector<int> numbers { 10, 20, 30, 40, 50 };

    // Display the vector elements.
    for (int val : numbers)
        cout << val << endl;

    return 0;
}
```

Program Output

```
10
20
30
40
50
```

Other Useful Member Functions

Member Function	Description	Example
`at(i)`	Returns the value of the element at position `i` in the vector	`cout << vec1.at(i);`
`capacity()`	Returns the maximum number of elements a vector can store without allocating more memory	`maxElements = vec1.capacity();`
`reverse()`	Reverse the order of the elements in a vector	`vec1.reverse();`
`resize (n, val)`	Resizes the vector so it contains `n` elements. If new elements are added, they are initialized to `val`.	`vec1.resize(5, 0);`
`swap(vec2)`	Exchange the contents of two vectors	`vec1.swap(vec2);`

A Two-dimensional Array in Program 7-21

Program 7-21

```
// This program demonstrates a two-dimensional array.
#include <iostream>
#include <iomanip>
using namespace std;

int main()
{
   const int NUM_DIVS = 3;            // Number of divisions
   const int NUM_QTRS = 4;            // Number of quarters
   double sales[NUM_DIVS][NUM_QTRS]; // Array with 3 rows and 4 columns.
   double totalSales = 0;             // To hold the total sales.
   int div, qtr;                      // Loop counters.

   cout << "This program will calculate the total sales of\n";
   cout << "all the company's divisions.\n";
   cout << "Enter the following sales information:\n\n";
```

(program continues)

A Two-dimensional Array in Program 7-21

Program 7-21 *(continued)*

```
   // Nested loops to fill the array with quarterly
   // sales figures for each division.
   for (div = 0; div < NUM_DIVS; div++)
   {
      for (qtr = 0; qtr < NUM_QTRS; qtr++)
      {
         cout << "Division " << (div + 1);
         cout << ", Quarter " << (qtr + 1) << ": $";
         cin >> sales[div][qtr];
      }
      cout << endl; // Print blank line.
   }

   // Nested loops used to add all the elements.
   for (div = 0; div < NUM_DIVS; div++)
   {
      for (qtr = 0; qtr < NUM_QTRS; qtr++)
         totalSales += sales[div][qtr];
   }

   cout << fixed << showpoint << setprecision(2);
   cout << "The total sales for the company are: $";
   cout << totalSales << endl;
   return 0;
}
```

A Two-dimensional Array in Program 7-21

Program Output with Example Input Shown in Bold

```
This program will calculate the total sales of
all the company's divisions.
Enter the following sales data:

Division 1, Quarter 1: $31569.45 [Enter]
Division 1, Quarter 2: $29654.23 [Enter]
Division 1, Quarter 3: $32982.54 [Enter]
Division 1, Quarter 4: $39651.21 [Enter]

Division 2, Quarter 1: $56321.02 [Enter]
Division 2, Quarter 2: $54128.63 [Enter]
Division 2, Quarter 3: $41235.85 [Enter]
Division 2, Quarter 4: $54652.33 [Enter]

Division 3, Quarter 1: $29654.35 [Enter]
Division 3, Quarter 2: $28963.32 [Enter]
Division 3, Quarter 3: $25353.55 [Enter]
Division 3, Quarter 4: $32615.88 [Enter]

The total sales for the company are: $456782.34
```